

| $n = 100$  |         |                 |                 |                 |                 |     |         |                 |                 |                 |                 |  |  |
|------------|---------|-----------------|-----------------|-----------------|-----------------|-----|---------|-----------------|-----------------|-----------------|-----------------|--|--|
| $r$        | Weight  | B(2, 2)         | B(3, 3)         | B(2, 5)         | B(5, 2)         | $r$ | Weight  | B(2, 2)         | B(3, 3)         | B(2, 5)         | B(5, 2)         |  |  |
| 2          | Uniform | 0.0186 (0.0019) | 0.0106 (0.0014) | 0.0036 (0.0008) | 0.0018 (0.0006) | 2   | Uniform | 0.0042 (0.0009) | 0.0012 (0.0005) | 0.0000 (0.0000) | 0.0000 (0.0000) |  |  |
| 2          | Central | 0.0314 (0.0025) | 0.0302 (0.0024) | 0.0170 (0.0018) | 0.0174 (0.0018) | 2   | Central | 0.0086 (0.0013) | 0.0032 (0.0008) | 0.0070 (0.0012) | 0.0052 (0.0010) |  |  |
| 2          | Tail    | 0.0166 (0.0018) | 0.0040 (0.0009) | 0.0016 (0.0006) | 0.0018 (0.0006) | 2   | Tail    | 0.0220 (0.0021) | 0.0038 (0.0009) | 0.0000 (0.0000) | 0.0000 (0.0000) |  |  |
| 5          | Uniform | 0.0310 (0.0025) | 0.0192 (0.0019) | 0.0122 (0.0016) | 0.0108 (0.0015) | 5   | Uniform | 0.0102 (0.0014) | 0.0066 (0.0011) | 0.0068 (0.0012) | 0.0074 (0.0012) |  |  |
| 5          | Central | 0.0420 (0.0028) | 0.0298 (0.0024) | 0.0326 (0.0025) | 0.0332 (0.0025) | 5   | Central | 0.0064 (0.0011) | 0.0038 (0.0009) | 0.0236 (0.0021) | 0.0200 (0.0020) |  |  |
| 5          | Tail    | 0.0270 (0.0023) | 0.0172 (0.0018) | 0.0024 (0.0007) | 0.0020 (0.0006) | 5   | Tail    | 0.0242 (0.0022) | 0.0154 (0.0017) | 0.0000 (0.0000) | 0.0000 (0.0000) |  |  |
| 8          | Uniform | 0.0380 (0.0027) | 0.0276 (0.0023) | 0.0114 (0.0015) | 0.0128 (0.0016) | 8   | Uniform | 0.0132 (0.0016) | 0.0074 (0.0012) | 0.0088 (0.0013) | 0.0090 (0.0013) |  |  |
| 8          | Central | 0.0392 (0.0027) | 0.0298 (0.0024) | 0.0244 (0.0022) | 0.0298 (0.0024) | 8   | Central | 0.0094 (0.0014) | 0.0062 (0.0011) | 0.0198 (0.0020) | 0.0200 (0.0020) |  |  |
| 8          | Tail    | 0.0362 (0.0026) | 0.0192 (0.0019) | 0.0028 (0.0007) | 0.0046 (0.0010) | 8   | Tail    | 0.0262 (0.0023) | 0.0192 (0.0019) | 0.0000 (0.0000) | 0.0004 (0.0003) |  |  |
| $n = 200$  |         |                 |                 |                 |                 |     |         |                 |                 |                 |                 |  |  |
| $r$        | Weight  | B(2, 2)         | B(3, 3)         | B(2, 5)         | B(5, 2)         | $r$ | Weight  | B(2, 2)         | B(3, 3)         | B(2, 5)         | B(5, 2)         |  |  |
| 2          | Uniform | 0.0002 (0.0002) | 0.0000 (0.0000) | 0.0000 (0.0000) | 0.0000 (0.0000) | 2   | Uniform | 0.0000 (0.0000) | 0.0000 (0.0000) | 0.0000 (0.0000) | 0.0000 (0.0000) |  |  |
| 2          | Central | 0.0002 (0.0002) | 0.0000 (0.0000) | 0.0018 (0.0006) | 0.0016 (0.0006) | 2   | Central | 0.0000 (0.0000) | 0.0000 (0.0000) | 0.0000 (0.0000) | 0.0000 (0.0000) |  |  |
| 2          | Tail    | 0.0354 (0.0026) | 0.0110 (0.0015) | 0.0000 (0.0000) | 0.0000 (0.0000) | 2   | Tail    | 0.0818 (0.0039) | 0.0266 (0.0023) | 0.0002 (0.0002) | 0.0000 (0.0000) |  |  |
| 5          | Uniform | 0.0012 (0.0005) | 0.0008 (0.0004) | 0.0056 (0.0011) | 0.0044 (0.0009) | 5   | Uniform | 0.0000 (0.0000) | 0.0000 (0.0000) | 0.0032 (0.0008) | 0.0028 (0.0007) |  |  |
| 5          | Central | 0.0000 (0.0000) | 0.0002 (0.0002) | 0.0132 (0.0016) | 0.0136 (0.0016) | 5   | Central | 0.0000 (0.0000) | 0.0000 (0.0000) | 0.0094 (0.0014) | 0.0046 (0.0010) |  |  |
| 5          | Tail    | 0.0216 (0.0021) | 0.0148 (0.0017) | 0.0004 (0.0003) | 0.0004 (0.0003) | 5   | Tail    | 0.0136 (0.0016) | 0.0162 (0.0018) | 0.0020 (0.0006) | 0.0032 (0.0008) |  |  |
| 8          | Uniform | 0.0006 (0.0003) | 0.0014 (0.0005) | 0.0070 (0.0012) | 0.0074 (0.0012) | 8   | Uniform | 0.0002 (0.0002) | 0.0000 (0.0000) | 0.0032 (0.0008) | 0.0036 (0.0008) |  |  |
| 8          | Central | 0.0016 (0.0006) | 0.0000 (0.0000) | 0.0146 (0.0017) | 0.0158 (0.0018) | 8   | Central | 0.0000 (0.0000) | 0.0000 (0.0000) | 0.0070 (0.0012) | 0.0066 (0.0011) |  |  |
| 8          | Tail    | 0.0126 (0.0016) | 0.0132 (0.0016) | 0.0006 (0.0003) | 0.0002 (0.0002) | 8   | Tail    | 0.0046 (0.0010) | 0.0088 (0.0013) | 0.0016 (0.0006) | 0.0016 (0.0006) |  |  |
| $n = 1000$ |         |                 |                 |                 |                 |     |         |                 |                 |                 |                 |  |  |
| $r$        | Weight  | B(2, 2)         | B(3, 3)         | B(2, 5)         | B(5, 2)         |     |         |                 |                 |                 |                 |  |  |
| 2          | Uniform | 0.0000 (0.0000) | 0.0000 (0.0000) | 0.0000 (0.0000) | 0.0000 (0.0000) |     |         |                 |                 |                 |                 |  |  |
| 2          | Central | 0.0000 (0.0000) | 0.0000 (0.0000) | 0.0000 (0.0000) | 0.0000 (0.0000) |     |         |                 |                 |                 |                 |  |  |
| 2          | Tail    | 0.1032 (0.0043) | 0.0446 (0.0029) | 0.0006 (0.0003) | 0.0010 (0.0004) |     |         |                 |                 |                 |                 |  |  |
| 5          | Uniform | 0.0000 (0.0000) | 0.0000 (0.0000) | 0.0032 (0.0008) | 0.0028 (0.0007) |     |         |                 |                 |                 |                 |  |  |
| 5          | Central | 0.0000 (0.0000) | 0.0000 (0.0000) | 0.0012 (0.0005) | 0.0004 (0.0003) |     |         |                 |                 |                 |                 |  |  |
| 5          | Tail    | 0.0108 (0.0015) | 0.0128 (0.0016) | 0.0054 (0.0010) | 0.0042 (0.0009) |     |         |                 |                 |                 |                 |  |  |
| 8          | Uniform | 0.0000 (0.0000) | 0.0000 (0.0000) | 0.0034 (0.0008) | 0.0050 (0.0010) |     |         |                 |                 |                 |                 |  |  |
| 8          | Central | 0.0000 (0.0000) | 0.0000 (0.0000) | 0.0044 (0.0009) | 0.0028 (0.0007) |     |         |                 |                 |                 |                 |  |  |
| 8          | Tail    | 0.0032 (0.0008) | 0.0078 (0.0012) | 0.0028 (0.0007) | 0.0036 (0.0008) |     |         |                 |                 |                 |                 |  |  |

Table 6: Simulation II: Empirical power and Monte Carlo standard error for different sample sizes, spacing orders, weighting schemes, and Beta alternatives. Entries are empirical power with Monte Carlo standard error in parentheses. The test is calibrated at  $\alpha = 0.05$  under  $U(0, 1)$ .